- long options pay theta,
- short options receive theta,
- ATM options usually have the largest magnitude of theta,
- theta usually accelerates near expiry for ATM options.

This is why traders often talk about whether a position is *cheap* or *expensive* in theta terms. A position with a lot of gamma but too much theta may still be unattractive if realized movement is unlikely to justify the carry cost.

17.10 Gamma-Theta Efficiency

Traders often compare a portfolio's gamma to the theta they are paying. Informally, they ask whether the gamma/theta relationship is *efficient*.

A common rule-of-thumb benchmark is:

$$\text{fair theta} \approx \frac{\text{cash gamma} \times \sigma^2}{200},$$

where σ is the daily implied move in percent.

Example 17.3 (Gamma-theta efficiency)**.** Suppose a trader has cash gamma

$$1{,}000{,}000$$

and implied volatility corresponds to a 1% expected daily standard deviation move.

Then the rough fair theta is

$$\frac{1{,}000{,}000 \times 1^2}{200} = 5{,}000.$$

So if the trader is paying \$8,000 of theta for that 1,000,000 of cash gamma, the position may be viewed as somewhat inefficient.

Remark 17.9. This benchmark is only a heuristic, but it captures a market-making instinct: gamma is not judged in isolation. It is judged relative to what it costs to own.

17.11 Why Theta Matters So Much

Theta matters because it is always running in the background. Even when the market does nothing, time is still passing.

So theta answers a basic question:

What is the expected carry cost of waiting?

If you are long options, doing nothing is costly. If you are short options, doing nothing is often beneficial.

That is why theta is one of the most operationally important Greeks in real books. It is not just a theoretical sensitivity. It is the daily bleed or daily earn of the portfolio.

18 Volatility

18.1 Why Volatility Matters

Before moving to vega, we need to define volatility clearly. Volatility is central to options because an option trader does not only care about *direction*; they also care about the *speed* and *magnitude* of the underlying's movement. In practice, option prices are closely tied to how much the market has moved or is expected to move.

Definition 18.1 (Volatility)**. Volatility** is the degree of variation of an underlying's price over time, typically measured as the standard deviation of logarithmic returns and quoted on an annualized basis.

A higher volatility means a wider distribution of possible future prices. A lower volatility means a tighter distribution concentrated closer to the current forward price.

18.2 Normal vs. Lognormal Thinking

Although many people first learn volatility through the normal distribution, option pricing usually works with a *lognormal* framework for the underlying price. This is because prices compound over time, and because stocks and futures cannot go below zero.

Remark 18.1. A lognormal distribution has a tighter downside and a longer upside tail. This better reflects how asset prices behave in practice.

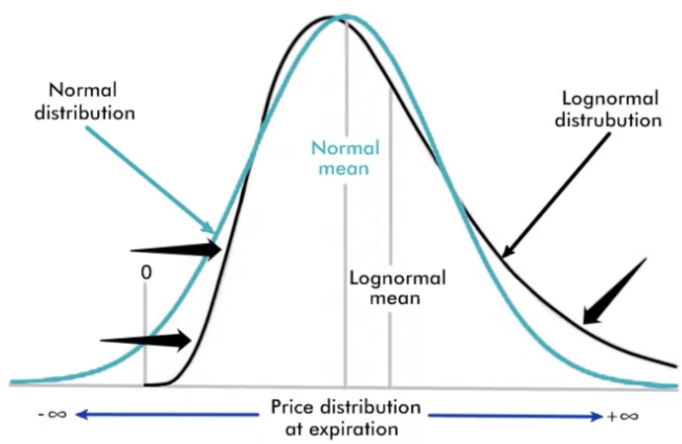

Figure 43: Normal versus lognormal distributions. In options pricing, the lognormal shape is preferred because prices compound and cannot fall below zero.

18.3 Volatility as a Distribution of Future Prices

Volatility can be interpreted as a statement about the distribution of future prices. For example, if an underlying is trading at 100 and has 20% annual volatility, then roughly speaking:

$$100 \pm 20$$

gives the one-year one-standard-deviation range, so the asset is expected to lie between 80 and 120 about 68% of the time.

Likewise, two standard deviations would give:

$$100 \pm 40,$$

so the range 60 to 140 contains roughly 95% of outcomes.

Remark 18.2. These percentages come from the usual one-standard-deviation and two-standard-deviation interpretation. In practice, the exact asset-price model is lognormal, so this is an approximation, but it is still very useful intuition.

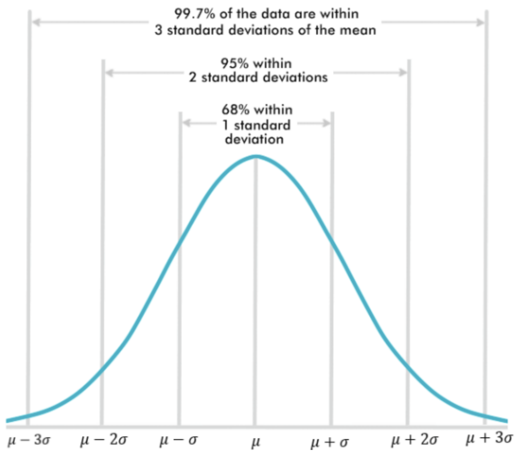

Figure 44: Standard-deviation interpretation of a probability distribution. Volatility is often understood through these one-, two-, and three-standard-deviation ranges.

Different volatility assumptions produce different price distributions. Lower volatility gives a tighter, taller distribution. Higher volatility gives a wider, flatter one.

Figure 45: Different volatility assumptions lead to different future price distributions. Higher volatility spreads probability mass over a wider range of outcomes.

18.4 Types of Volatility

There are several different volatility concepts used by traders.

Definition 18.2 (Historical Volatility)**. Historical volatility** (or **realized volatility**) measures how much the underlying has moved over a past period using observed price data.

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A useful practical distinction is this:

- **Historical / realized volatility** is backward-looking.
- **Implied volatility** is forward-looking.

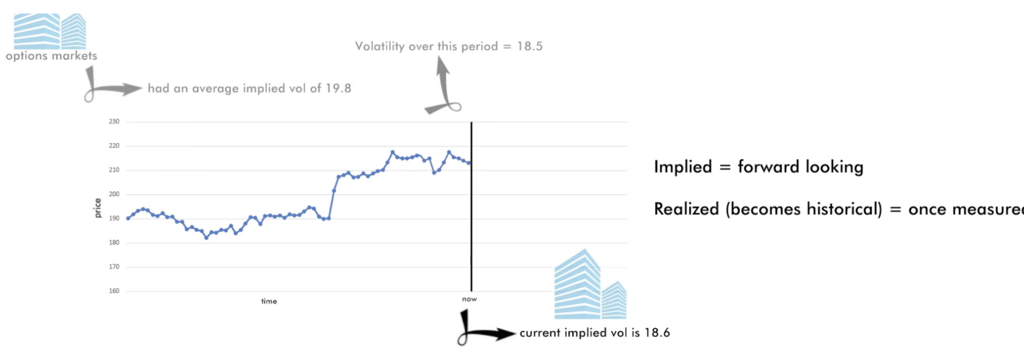

Figure 46: Implied volatility is forward-looking, while realized volatility becomes historical once the movement has already occurred.

18.5 Annualized Volatility

Volatility is usually quoted as an annualized number. If σ_{daily} is the standard deviation of daily log returns, then annualized volatility is approximately:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}.$$

More generally, if σ_{annual} is annualized volatility, then the one-standard-deviation move over t trading days is:

$$\sigma_t = \sigma_{\text{annual}}\sqrt{\frac{t}{252}}.$$

Generalize, annualized, volatility for a time horizon, t, in years is expressed as:

$$\sigma_t = \sigma_{annually}\sqrt{t}$$

So, if we have a daily standard deviation of logarithmic returns, we can find the annualized vol as:

$$\sigma_{annually} = \sigma_{daily}\sqrt{t}$$

252 trading days per year

1% daily standard deviation = 0.01

$$= 0.01\sqrt{252} = 0.1587$$

So, we say that the historical volatility over this period was a 15.87

Or, more generally, we express:

$$Expected\ 1\ s.d.move\ over\ t\ days = \sigma_{annually}\sqrt{\frac{t}{252}}$$

N.B. this is not to be confused with the expected daily move, which is given by:

$$Expected\ daily\ move = \frac{\sigma_{annually}}{\sqrt{\frac{t}{252}}} \times \sqrt{\frac{2}{\pi}}$$

Figure 47: Converting between daily and annualized volatility, and using volatility to estimate the one-standard-deviation move over a horizon of t trading days.

Example 18.1 (From annual volatility to a one-day standard deviation move)**.** Suppose volatility is 20%, or 0.20 in decimal form. Then the one-day standard deviation move is

$$0.20\sqrt{\frac{1}{252}} \approx 0.0126,$$

which is about 1.26%.

18.6 Standard Deviation Move vs. Expected Move

A very important distinction is that a one-standard-deviation move is *not* the same thing as the expected daily move.

For example, a 16 vol product has approximately a 1% one-day standard deviation move, since

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But its expected daily move is smaller, about 0.8%, not 1%.

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They are related, but they are not the same.

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The logic is straightforward:

- square each daily log return,
- add them up,
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- multiply by 252 to annualize,
- and then take the square root.

Remark 18.4. This is slightly different from the way standard deviation is often introduced in statistics courses, which is why the lecture explicitly points it out.

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18.9 Time-Variance Additivity

Another important fact is that **variance**, not volatility itself, adds through time.

So if we know the annualized vol over one 10-day period and over the next 10-day period, the 20-day vol is *not* the simple average of those two vols.

Instead, the correct way is to combine the variances and then take a square root at the end.

Remark 18.5. This is why traders often say that *time-variance is additive.* Volatility itself is not linearly additive.

18.10 Why Volatility Comes Before Vega

This section comes right before vega for a reason. Vega measures how much an option price changes when implied volatility changes. So before discussing vega, we need a clear picture of what volatility itself means.

Volatility is not just an abstract statistic. It is the language traders use to describe how wide the future price distribution is, how much movement the market is pricing, and how expensive optionality should be.

Remark 18.6. A good practical summary is this: historical vol tells us how much the market *did* move, while implied vol tells us how much the market is *pricing* that it may move.

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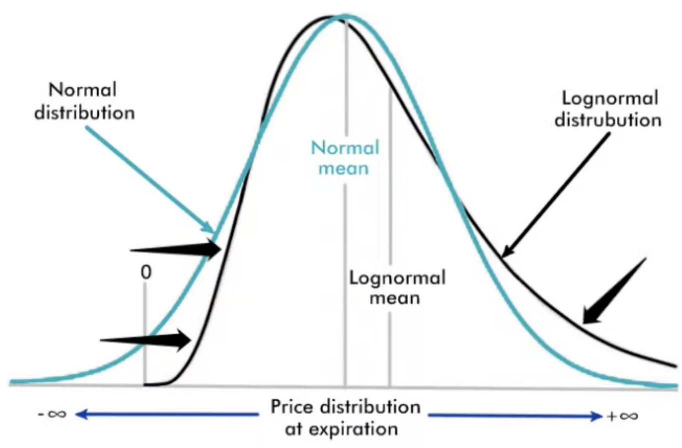

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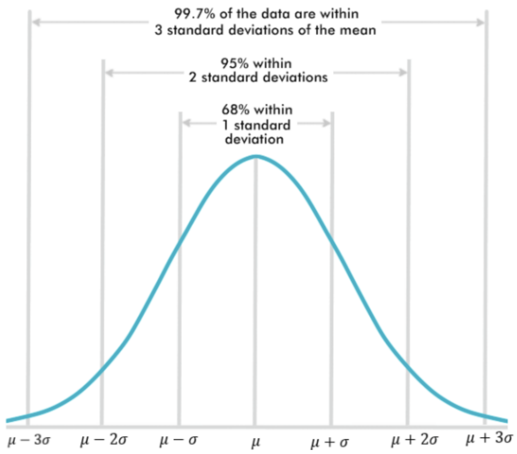

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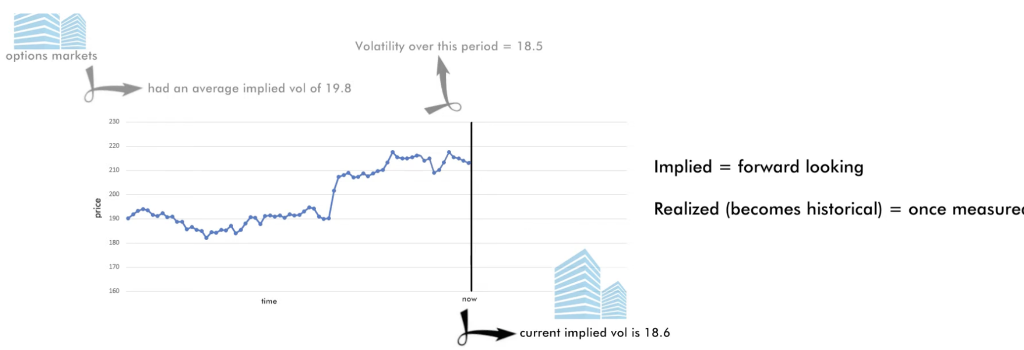

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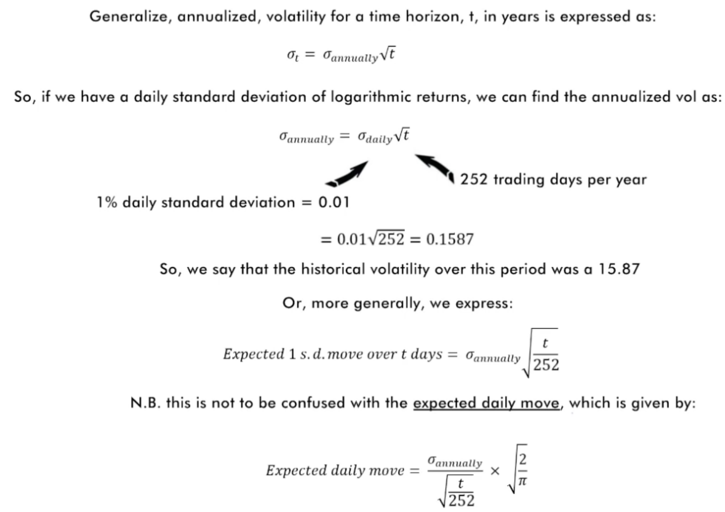

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19.11 Vega

Definition 19.6 (Vega)**.** The **vega** of an option is the change in the option's value for a one-point change in implied volatility.

Vega measures sensitivity to implied volatility, just as delta measures sensitivity to the underlying price. Since options traders constantly translate prices into volatility terms, vega is one of the most watched Greeks on an options desk.

Remark 19.7. Vega is quoted per 1-point move in implied volatility. So if an option has vega 0.081, then increasing implied volatility from 23.52 to 24.52 would increase the option's value by about 0.081.

Example 19.2 (Single-option vega change)**.** Suppose a call option has:

$$\text{price} = 1.00, \qquad \text{delta} = 0.30, \qquad \text{vega} = 0.04, \qquad \text{IV} = 15.$$

If implied volatility increases from 15 to 16, then the option price increases by:

$$0.04 \times 1 = 0.04.$$

So the new option price is approximately

$$1.00 + 0.04 = 1.04.$$

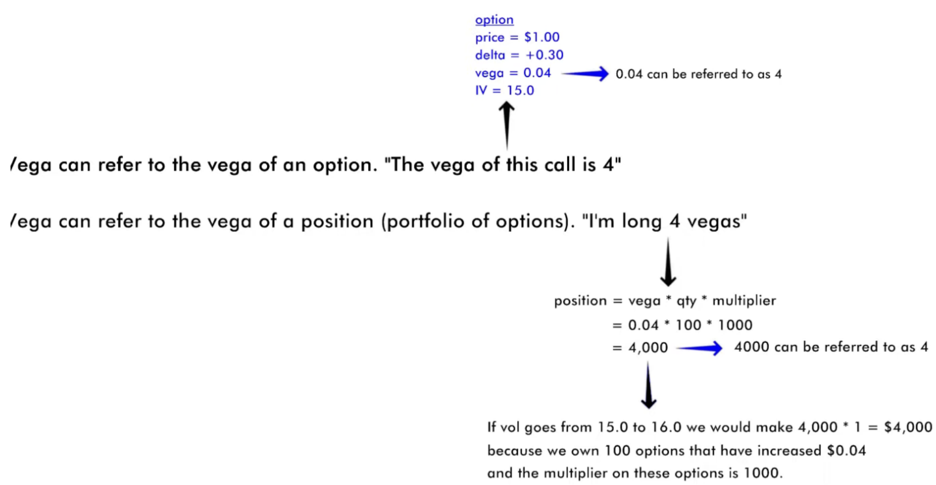
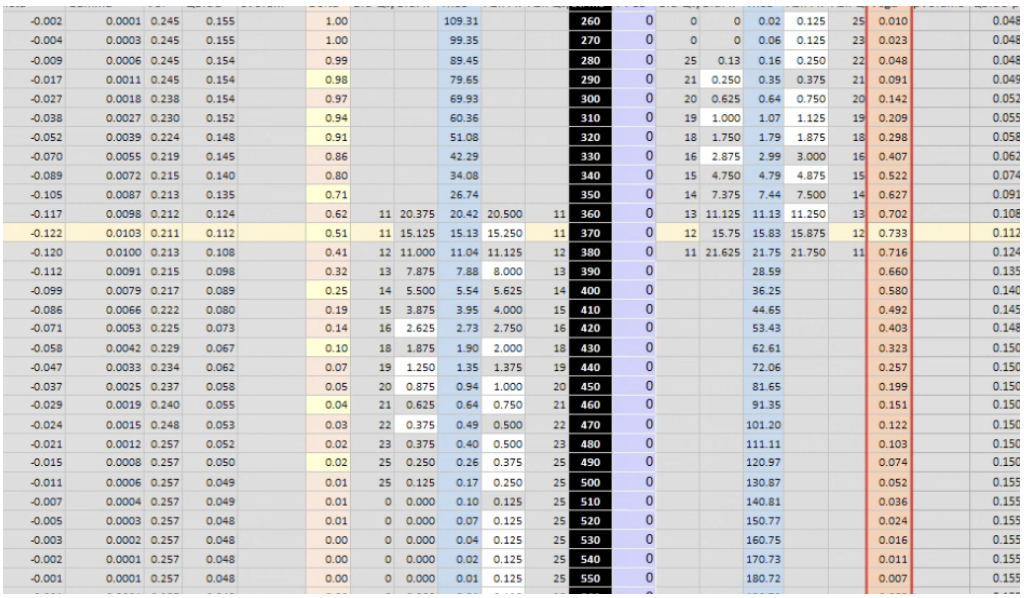


Figure 53: Vega can refer either to the vega of a single option or to the vega of an entire options position.

Vega on the Options Board

On the options board, calls and puts on the same strike share the same vega. This is another reflection of put-call parity intuition: at a fixed strike, the call and put respond the same way to a volatility change.

For example, on the highlighted strike, the call and put both have vega 0.081. If implied volatility rises by one point, each option on that line gains 0.081 in value.



-0.002	0.0001	0.245	0.155		1.00			109.31			260	0	0	0	0.02	0.125	25	0.010		0.048
-0.004	0.0003	0.245	0.155		1.00			99.35			270	0	0	0	0.06	0.125	23	0.023		0.048
-0.009	0.0006	0.245	0.154		0.99			89.45			280	0	25	0.13	0.16	0.250	22	0.048		0.048
-0.017	0.0011	0.245	0.154		0.98			79.65			290	0	21	0.250	0.35	0.375	21	0.091		0.049
-0.027	0.0018	0.238	0.154		0.97			69.93			300	0	20	0.625	0.64	0.750	20	0.142		0.052
-0.038	0.0027	0.230	0.152		0.94			60.36			310	0	19	1.000	1.07	1.125	19	0.209		0.055
-0.052	0.0039	0.224	0.148		0.91			51.08			320	0	18	1.750	1.79	1.875	18	0.298		0.058
-0.070	0.0055	0.219	0.145		0.86			42.29			330	0	16	2.875	2.99	3.000	16	0.407		0.062
-0.089	0.0072	0.215	0.140		0.80			34.08			340	0	15	4.750	4.79	4.875	15	0.522		0.074
-0.105	0.0087	0.213	0.135		0.71			26.74			350	0	14	7.375	7.44	7.500	14	0.627		0.091
-0.117	0.0098	0.212	0.124		0.62	11	20.375	20.42	20.500	11	360	0	13	11.125	11.13	11.250	13	0.702		0.108
-0.122	0.0103	0.211	0.112		0.51	11	15.125	15.13	15.250	11	370	0	12	15.75	15.83	15.875	12	0.733		0.112
-0.120	0.0100	0.213	0.108		0.41	12	11.000	11.04	11.125	12	380	0	11	21.625	21.75	21.750	11	0.716		0.124
-0.112	0.0091	0.215	0.098		0.32	13	7.875	7.88	8.000	13	390	0			28.59			0.660		0.135
-0.099	0.0079	0.217	0.089		0.25	14	5.500	5.54	5.625	14	400	0			36.25			0.580		0.140
-0.086	0.0066	0.222	0.080		0.19	15	3.875	3.95	4.000	15	410	0			44.65			0.492		0.145
-0.071	0.0053	0.225	0.073		0.14	16	2.625	2.73	2.750	16	420	0			53.43			0.403		0.148
-0.058	0.0042	0.229	0.067		0.10	18	1.875	1.90	2.000	18	430	0			62.61			0.323		0.150
-0.047	0.0033	0.234	0.062		0.07	19	1.250	1.35	1.375	19	440	0			72.06			0.257		0.150
-0.037	0.0025	0.237	0.058		0.05	20	0.875	0.94	1.000	20	450	0			81.65			0.199		0.150
-0.029	0.0019	0.240	0.055		0.04	21	0.625	0.64	0.750	21	460	0			91.35			0.151		0.150
-0.024	0.0015	0.248	0.053		0.03	22	0.375	0.49	0.500	22	470	0			101.20			0.122		0.150
-0.021	0.0012	0.257	0.052		0.02	23	0.375	0.40	0.500	23	480	0			111.11			0.103		0.150
-0.015	0.0008	0.257	0.050		0.02	25	0.250	0.26	0.375	25	490	0			120.97			0.074		0.150
-0.011	0.0006	0.257	0.049		0.01	25	0.125	0.17	0.250	25	500	0			130.87			0.052		0.155
-0.007	0.0004	0.257	0.049		0.01	0	0.000	0.10	0.125	25	510	0			140.81			0.036		0.155
-0.005	0.0003	0.257	0.048		0.01	0	0.000	0.07	0.125	25	520	0			150.77			0.024		0.155
-0.003	0.0002	0.257	0.048		0.00	0	0.000	0.04	0.125	25	530	0			160.75			0.016		0.155
-0.002	0.0001	0.257	0.048		0.00	0	0.000	0.02	0.125	25	540	0			170.73			0.011		0.155
-0.001	0.0001	0.257	0.048		0.00	0	0.000	0.01	0.125	25	550	0			180.72			0.007		0.155

Figure 54: Option board with the vega column highlighted. Calls and puts on the same strike have the same vega.

Long Options are Long Vega

Every outright option has positive vega. That means:

- buying a call gives positive vega,
- buying a put gives positive vega,
- selling an outright option gives negative vega.

So if implied volatility rises, long-option holders benefit because the options become more valuable. If implied volatility falls, they lose value.

Remark 19.8. This is why options traders are often called *volatility traders*. Even when they talk about price, they are often really thinking about where implied volatility should trade.

How Vega Changes Across Time to Expiry

Vega is not constant across expiries. Longer-dated options have larger vegas because there is more time for volatility to matter.

This means a one-point change in implied volatility has a larger price impact on a long-dated option than on a short-dated one.

How Vega Changes with Volatility

Vega also varies with the level of implied volatility itself. In the examples from the lesson, lowering the at-the-money volatility decreases vega away from the peak region.

So vega depends on at least three major things:

- time to expiry,

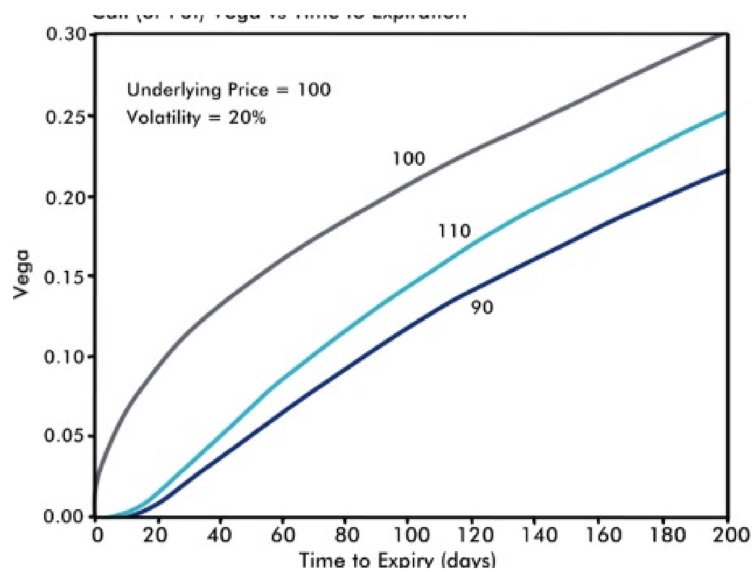


Figure 55: Vega versus time to expiry. For a fixed underlying and volatility, vega generally increases as time to expiry increases.

- distance from at the money,
- and the current volatility level.

Vega Across Strike Space

Vega is largest for options near the at-the-money strike. As we move further ITM or OTM, vega falls.

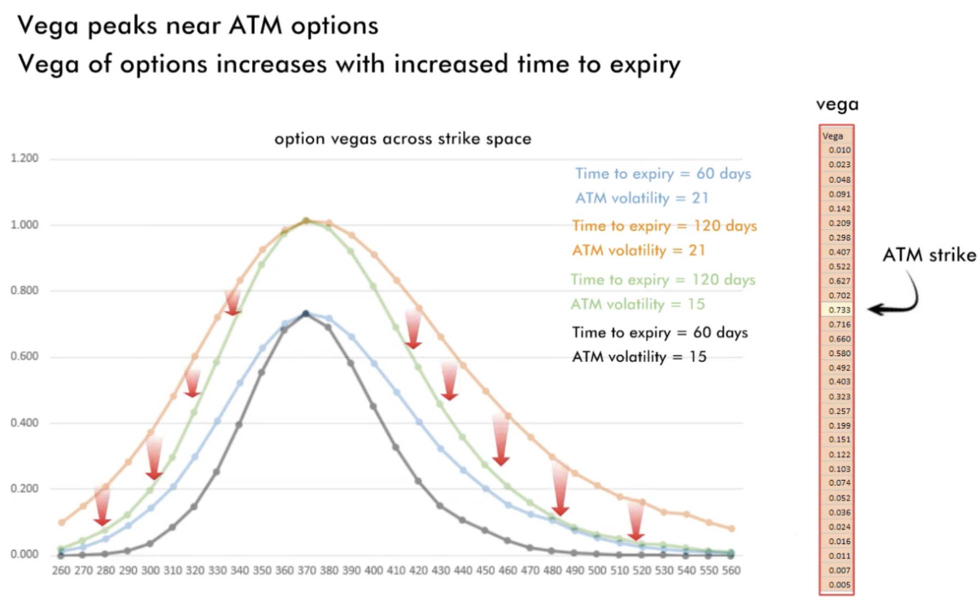


Figure 56: Vegas across strike space. Vega is highest near the at-the-money strike and declines as strikes move away from ATM.

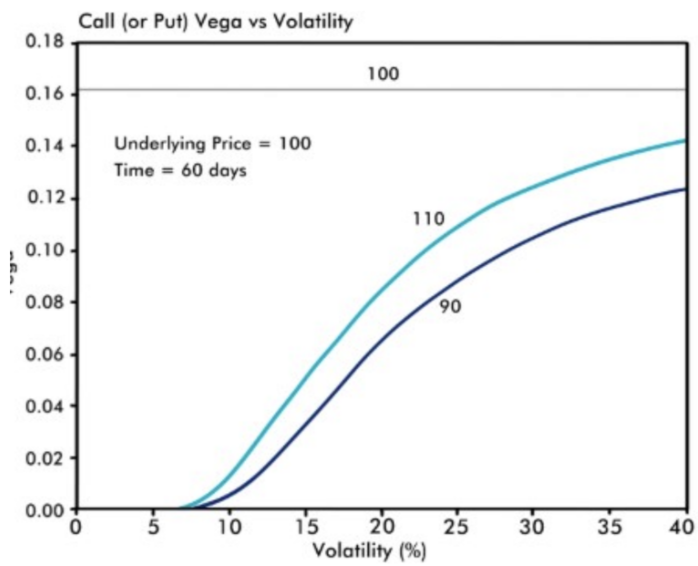


Figure 57: Vega versus implied volatility for different strikes. Vega changes with the volatility level itself.

Remark 19.9. Options closest to ATM are the most sensitive to a change in implied volatility. That is why the vega curve peaks near the current underlying level.

The 3D surface view shows the same pattern more globally: the vega ridge sits near ATM strikes and becomes larger for later expiries.

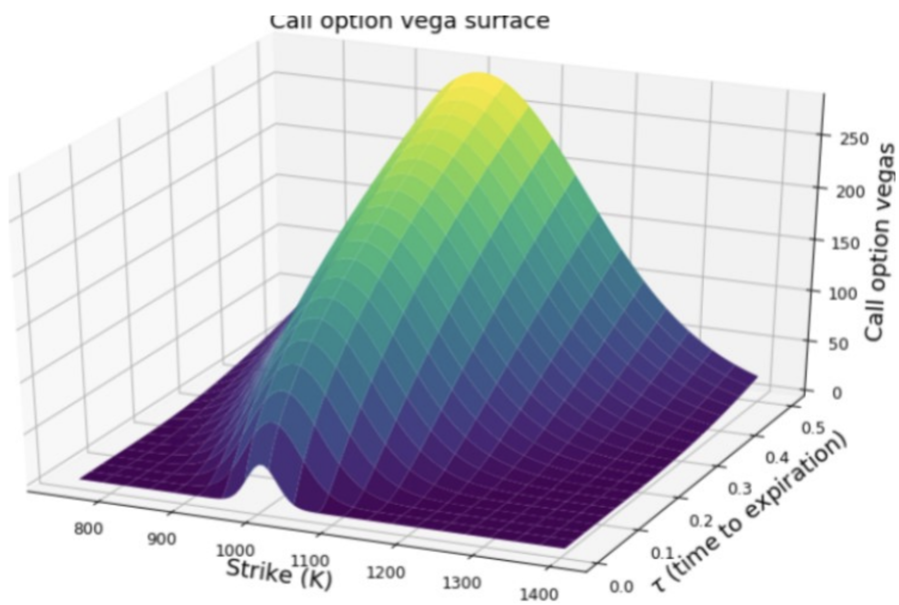


Figure 58: Vega surface across strike and time. Vega peaks near ATM strikes and rises with longer time to expiry.

Portfolio Vega

Like delta, traders also use the word *vega* to describe a whole position, not just a single option.

Definition 19.7 (Portfolio Vega)**.** The vega of an options portfolio is

$$\text{portfolio vega} = (\text{number of options}) \times (\text{multiplier}) \times (\text{option vega}).$$

Example 19.3 (Position vega)**.** Suppose we buy 1000 options, each with vega 0.081, and the multiplier is 50. Then:

$$\text{portfolio vega} = 1000 \times 50 \times 0.081 = 4050.$$

So a one-point increase in implied volatility produces about a

$$\$4050$$

gain, while a one-point decrease produces about a

$$\$4050$$

loss.

Traders might say, "I'm long 4050 vegas." In context, this refers to the vega of the *position*, not the per-option vega number.

Practical Vega Language

The word *vega* is used in two common ways:

- the vega of a single option, such as "this line has 0.081 vega,"
- the vega of a position, such as "I'm long 4050 vegas."

This can be confusing at first, but traders usually rely on context.

Why Vega Matters So Much

Vega is one of the most important Greeks because implied volatility is one of the main inputs that market makers actively control in their models. If a trader changes the implied volatility input, option theos move immediately. So understanding vega means understanding how volatility views turn into pricing changes.

In practice, vega tells us:

- how sensitive an option is to a volatility repricing,
- which strikes are most exposed to vol changes,
- which expiries carry more volatility risk,
- and how much a whole portfolio will gain or lose from an IV shift.

19.12 Rho & Boxes

Rho

Definition 19.8 (Rho)**.** The **rho** of an option is the sensitivity of that option's value to a change in interest rates. In other words, rho is the change in option price for a 1-point (1%) change in rates.

Rho is usually the least emphasized Greek in basic options discussions, but it still matters. In long-dated options, large books, or environments where rates are moving meaningfully, rho can become important very quickly. Unlike delta, gamma, theta, and vega, rho is tied to the interest-rate assumptions built into the pricing model.

Example 19.4 (Single-option rho)**.** Suppose an option has:

$$\text{theoretical value} = 0.93, \qquad \rho = 0.02, \qquad r = 3.5\%.$$

If the interest rate rises from 3.5% to 4.5%, then the option value increases by:

$$0.02 \times 1 = 0.02.$$

So the new theoretical value becomes:

$$0.93 + 0.02 = 0.95.$$

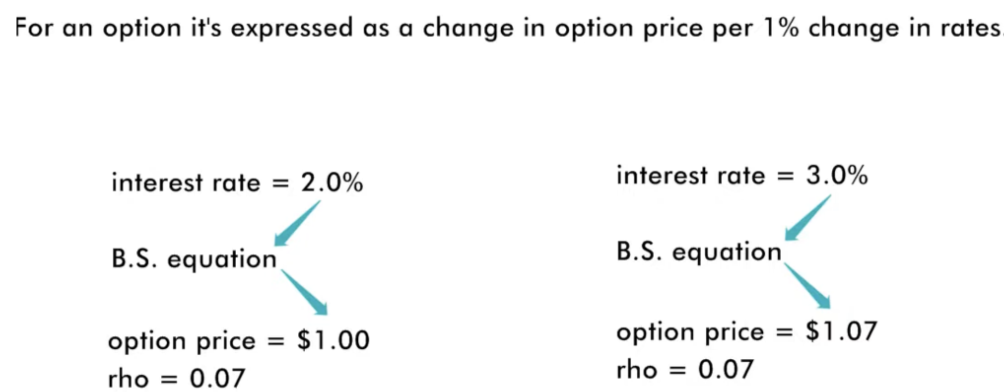


Figure 59: Rho for a single option is the change in option price for a 1% change in interest rates.

Remark 19.10. Unlike vega, calls and puts on the same strike do *not* necessarily have the same rho. Interest rates affect calls and puts differently.

Book Rho

In practice, market makers usually do not focus on rho one option at a time. Instead, they think about rho at the level of the whole book.

Definition 19.9 (Book rho)**.** The **rho of a portfolio** is the total P&L change of that portfolio for a 1% change in interest rates.

So if a desk says it is *long rho*, that means the book makes money when rates rise and loses money when rates fall. If the desk is *short rho*, the opposite is true.

Example 19.5 (Portfolio rho)**.** Suppose we own 200 options, each with rho 0.07, and the multiplier is 100. Then the book rho is:

$$200 \times 0.07 \times 100 = 1400.$$

So:

- if rates rise by 1%, the book gains about \$1400,
- if rates fall by 0.5%, the book loses about \$700.

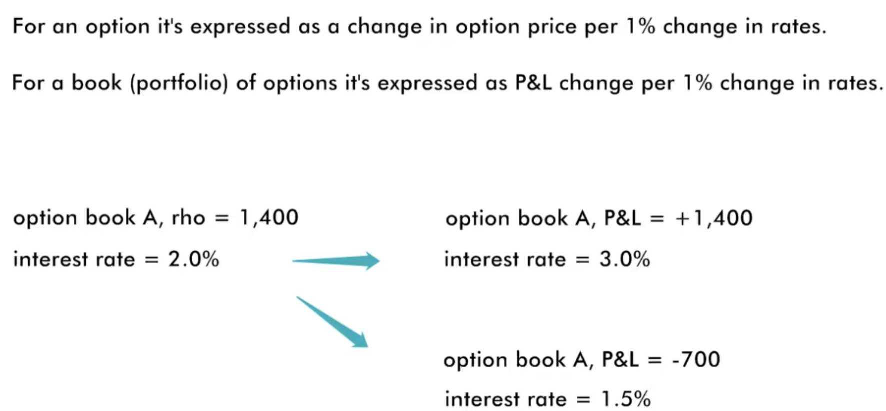


Figure 60: Book rho is expressed as total P&L change for a 1% move in rates.

Because large firms trade many products, rho can also be aggregated across desks and managed at the firm level.

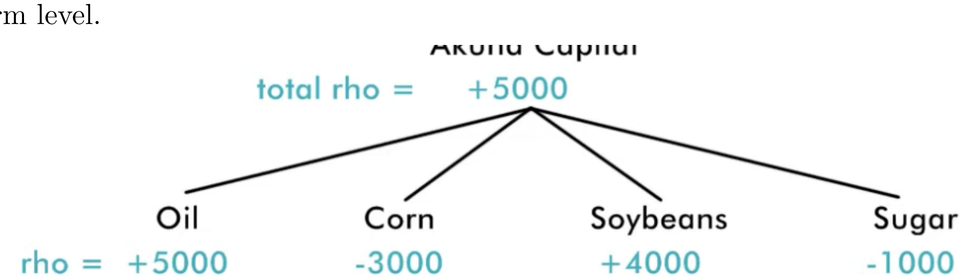


Figure 61: Rho can be summed across products and managed at the firm level.

Interest Rate Curves

There is not just one single interest rate plugged into an options model. Instead, traders use an **interest-rate curve**, which gives different modeled rates for different expiries.

Definition 19.10 (Interest-rate curve)**.** An **interest-rate curve** is the term structure of rates used across time in the options model. Different expiries can therefore use different effective rates.

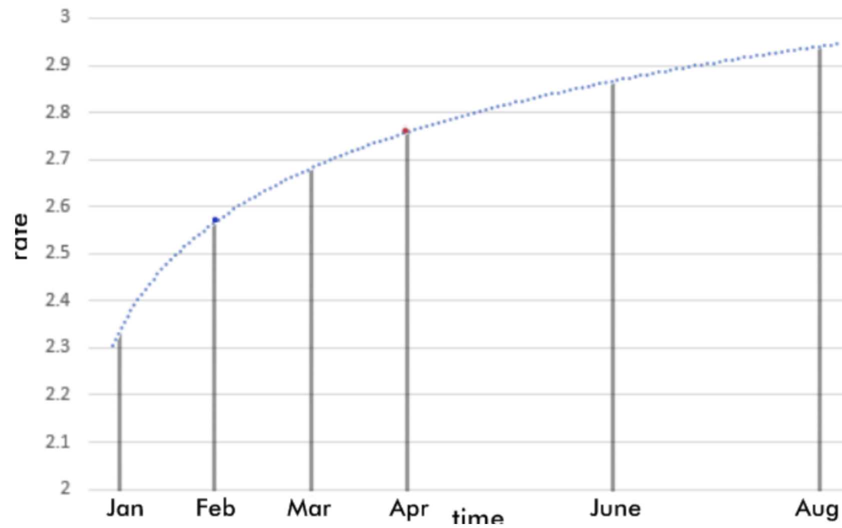


Figure 62: Example of an interest-rate curve across time. Different expiries can sit on different points of the curve.

This matters because a trader can be long rho in one expiry and short rho in another. In that setting, risk depends not only on whether the whole curve shifts up or down, but also on how its *shape* changes.

In practice, traders may think about being long rho on one point of the curve and short rho on another. That means the desk is exposed not just to level changes, but to curve-shape changes as well.

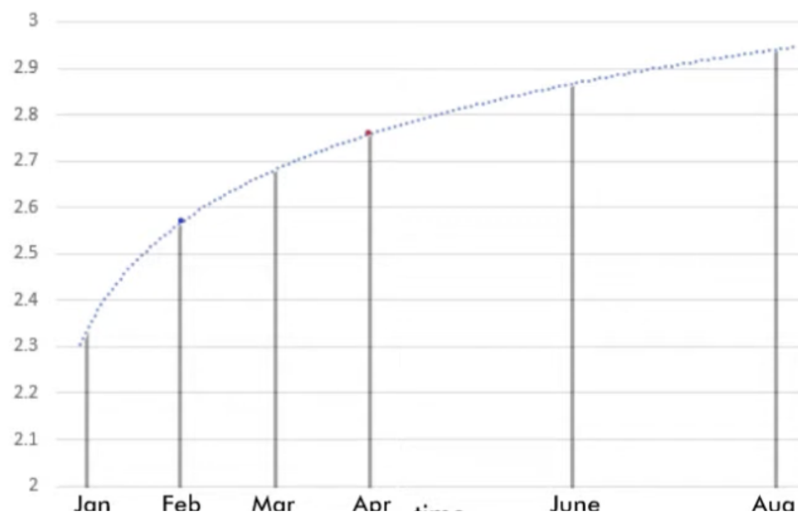


Figure 63: A trader can be long rho at one expiry and short rho at another, creating curve-shape exposure.

The benchmark rates underlying these models are tied to real economic rates, such as Fed Funds and related borrowing costs. When rates were near zero, rho often looked unimportant. As rates started moving again, rho became much more relevant to option desks.

Borrowing, Lending, and Rho Intuition

A useful way to build intuition is through borrowing and lending.

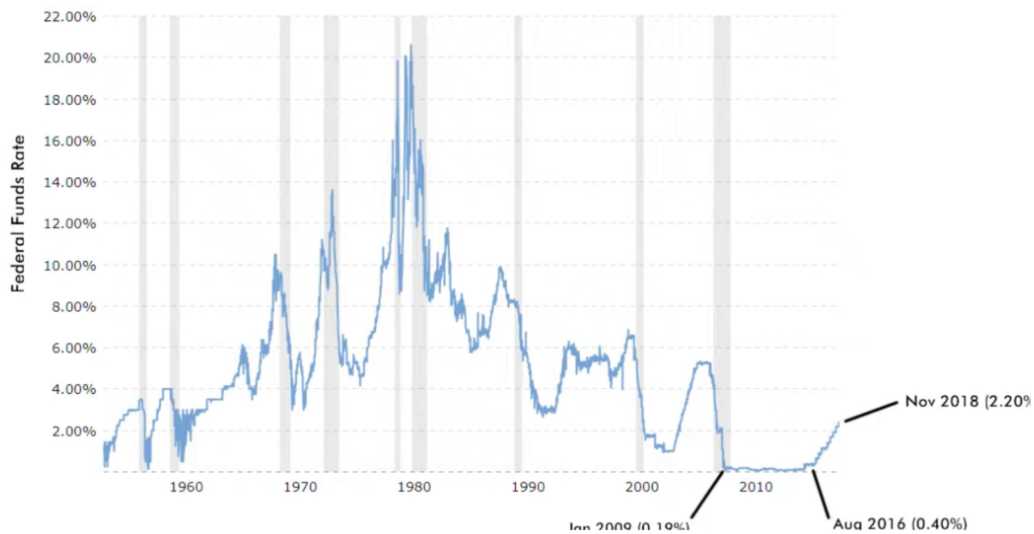


Figure 64: Historical interest-rate environments can materially change how important rho is in practice.

- If you are **borrowing** money, you generally benefit when rates rise after you lock in a lower rate.
- If you are **lending** money, you generally lose when rates rise, because you are stuck earning the old lower rate while new lending opportunities pay more.

This is why:

- a **long rho** position behaves like borrowing money,
- a **short rho** position behaves like lending money.

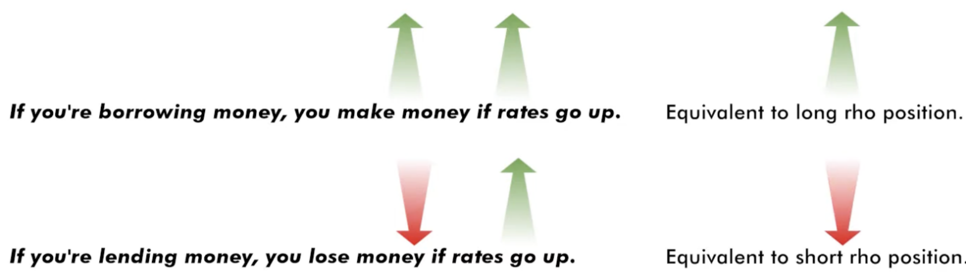


Figure 65: Long rho behaves like borrowing money; short rho behaves like lending money.

Boxes

A **box** is a four-leg options structure built from two strikes. It is one of the cleanest ways to see the connection between options and interest rates.

Definition 19.11 (Box spread)**.** A **box** is a combination of:

$$+K_1c - K_1p + K_2p - K_2c$$

across two strikes K_1 and K_2.

For example, a 700/1200 box is:

$$+700c - 700p + 1200p - 1200c.$$

At expiration, that structure is always worth:

$$1200 - 700 = 500,$$

regardless of where the future settles.

Remark 19.11. A box has a fixed payoff at expiry equal to the difference between the strikes. That is why it behaves like a financing instrument.

Why the Box is Not Worth 500 Today

If the box will definitely be worth 500 at expiry, it should still be worth *less* than 500 today because money has time value.

Suppose the model says the fair present value of the box is:

$$499.20.$$

Then the difference

$$500.00 - 499.20 = 0.80$$

is the financing value associated with tying up capital until expiry.

So the box is essentially a discounted fixed future cash flow.

Definition 19.12 (Present-value interpretation)**.** If a box pays a known amount at expiry, then its current value is the present value of that payoff under the modeled interest rate.

Who is Borrowing and Who is Lending?

If one trader **buys** the box today for 499.20 and receives 500.00 at expiry, that trader is effectively **lending** money.

If another trader **sells** the box today for 499.20 and must pay 500.00 at expiry, that trader is effectively **borrowing** money.

So:

- **buyer of the box** = lender,
- **seller of the box** = borrower.

How Rates Affect Box Prices

Because the box is a discounted fixed payoff:

- if interest rates go **up**, the present value of the future 500 goes **down**,

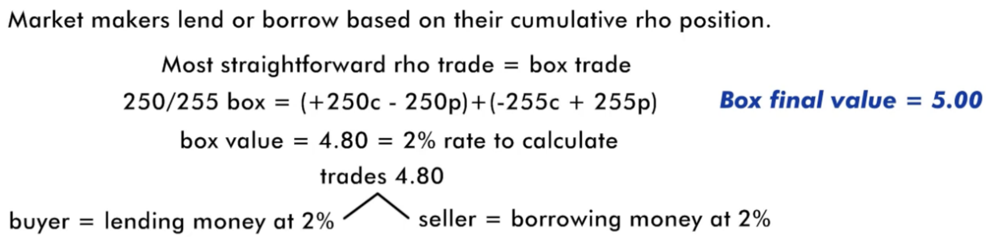


Figure 66: A box spread can be interpreted as lending or borrowing money through options.

- if interest rates go **down**, the present value of the future 500 goes **up**.

That means:

- higher rates push the box price lower,
- lower rates push the box price higher.

So the lender who bought the box prefers lower discounting after entry, while the borrower who sold the box prefers higher discounting after entry.

Backing Out the Implied Rate

If the final box value at expiry is known and the current box price is observed, then we can back out the implied financing rate from the present-value relation.

If B_0 is the current box price and B_T is the expiry payoff, then conceptually:

$$B_0 = \frac{B_T}{(1 + rT)}$$

under simple discounting, or more generally with the appropriate compounding convention.

So if:

$$B_T = 500, \qquad B_0 = 499.20,$$

then the implied rate is the rate that discounts 500 to 499.20 over the life of the box.

Remark 19.12. This is why traders can quote boxes either in price terms or in implied-rate terms. A box market is really also a funding market.

Why This Matters

Boxes are important because they isolate financing logic inside option structures. They show that options are not just about direction and volatility. They also encode interest-rate relationships.

That is why rho, even if often treated as the least important Greek, still matters:

- for long-dated options,
- for large portfolios,
- for funding-sensitive structures like boxes,
- and for firm-level risk management.

Closing Thoughts

Greeks vs. Payoff Diagrams

This course built a practical framework for thinking about how option models work and how market makers use them. A payoff diagram tells us what happens *at expiry*. Greeks tell us how an option behaves *before expiry*, while time, volatility, rates, and the underlying price are all still changing.

A useful way to close the course is to compare those two views directly. Consider a 100-strike call shown across several times to expiry. The expiry payoff is the familiar hockey-stick shape, but before expiry the option price is smoother and higher because there is still optionality left. This is exactly where the Greeks matter.

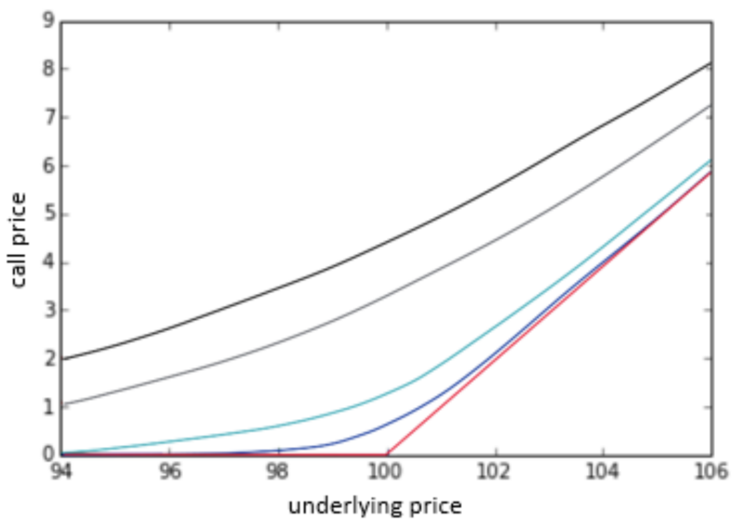


Figure 67: Call values across underlying prices for different times to expiry. The expiry payoff is the sharp red line, while pre-expiry curves remain smoother because they still contain time value.

From this picture, we can read several ideas at once:

- **Delta** is the local slope of each curve.
- **Gamma** is how quickly that slope changes.
- **Time value** is the gap between the expiry payoff and the earlier-dated curves.
- **ATM options** tend to have the most interesting behavior because that is where the curve bends the most.

Near expiry, the curve becomes steeper around the strike and flatter away from it. That is another way to see why ATM gamma becomes large as expiration approaches. A tiny move in the underlying can change the hedge ratio much more dramatically when the option is near the strike and near expiry.

Vega and theta are a little harder to read off a single price-versus-underlying plot, but the earlier sections showed how they fit into the same story. Vega measures how sensitive that curve is to changes in implied volatility. Theta measures how the curve decays as time passes. Together, the Greeks explain how the option moves from a smooth pre-expiry surface to the hard payoff shape at expiration.

The same idea can also be viewed as a surface over strike and time-to-expiry rather than a collection of slices.

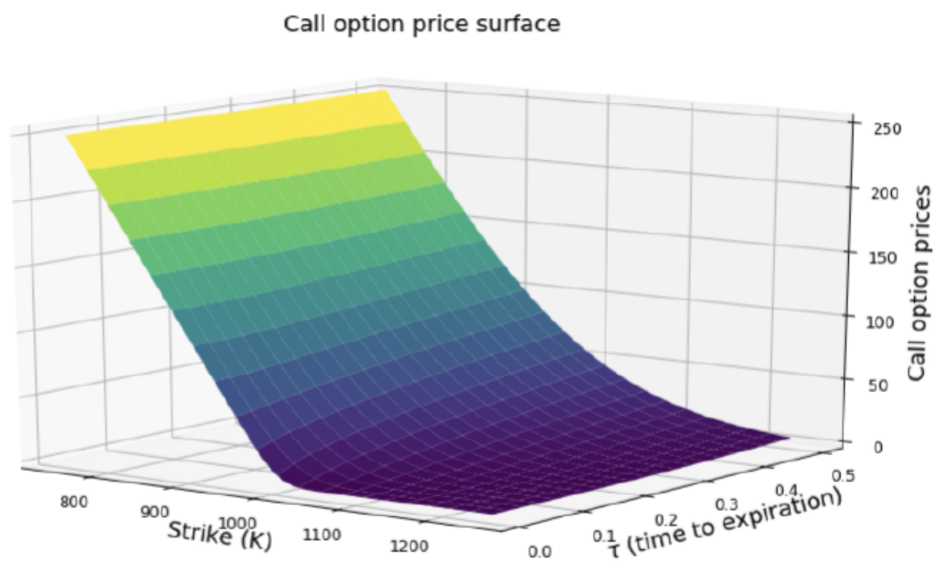


Figure 68: Option price surface across strike and time to expiry. This is another view of the same pricing relationships described by payoff diagrams and Greeks.

This surface perspective is useful because it emphasizes that an option price is not just one number. It is the output of a model taking several inputs at once. Changing the underlying, time, volatility, or rates moves us to a different point on that surface.

Models and Option Prices

For a market maker, theoretical values act as the anchor for quoting two-sided markets. They are not just academic outputs. They are the working midpoint around which bids and offers are made, risk is measured, and P&L is understood.

One of the most important habits in options trading is translating between **option prices** and **implied volatilities**. There are two equally valid ways to think about this process.

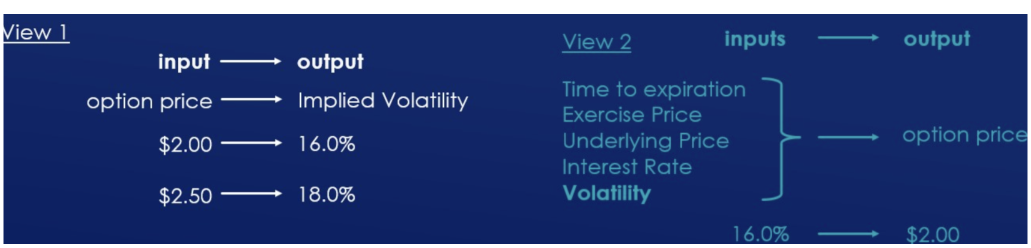


Figure 69: Two equivalent views: translating option prices into implied volatility, or inputting implied volatility into a model to produce an option price.

View 1: Price $\to$ Implied Volatility. If we observe a traded option price, we can back out the implied volatility that makes the pricing model produce that price. This is useful because volatility often gives a cleaner comparison across options than raw premium does. Option prices alone mix together strike, expiry, underlying level, and the rest of the model inputs. Implied volatility normalizes a lot of that information into a more comparable language.

View 2: Implied Volatility $\to$ Price. We can also start from a volatility assumption and feed it into the model along with strike, expiry, underlying price, and rates. The model then outputs an option value. This is the operational view used by market makers when they build quotes and manage risk.

Both views are correct. In practice, options desks constantly switch between them. Sometimes the trader thinks in volatility first and asks what price the model should produce. Other times the trader sees a price in the market and asks what volatility that price implies.

Remark 19.13. This is one of the deepest themes in options trading: option prices and implied volatilities are two languages describing the same object. Skilled traders become fluent in both.

What This Course Was Really About

At a high level, this course was not just about memorizing definitions like delta, gamma, theta, vega, and rho. It was about learning how option market makers think.

They think in terms of:

- theoretical values,
- fair markets around those values,
- sensitivities of price to model inputs,
- and how a portfolio behaves as those inputs move.

That is what turns a list of formulas into a trading framework.

The real market is dynamic. Underlyings move. Time passes. Volatility changes. Interest rates shift. A static payoff diagram only tells the final chapter. Greeks describe the whole story leading up to that ending.